From Classroom/Integrated School Programs 7 in Top 20, 23 in Top 100, 54 in Top 300, 106 in Top 500 All India Ranks & 2314 Students from Classroom /Integrated School Programs & 3723 Students from All Programs have been Awarded a Rank in JEE (Advanced), 2013

FIITJEE

ALL INDIA TEST SERIES

JEE(Main)-2014

ANSWERS, HINTS & SOLUTIONS
CRT–IV
(Main)

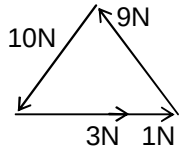
Q. No.	PHYSICS	CHEMISTRY	MATHEMATICS
1.	**A**	**D**	**D**
2.	**B**	**B**	**B**
3.	**D**	**C**	**C**
4.	**D**	**A**	**D**
5.	**C**	**B**	**B**
6.	**C**	**B**	**A**
7.	**A**	**D**	**C**
8.	**A**	**A**	**C**
9.	**B**	**B**	**C**
10.	**C**	**D**	**B**
11.	**A**	**A**	**A**
12.	**C**	**C**	**D**
13.	**D**	**C**	**A**
14.	**B**	**A**	**C**
15.	**A**	**A**	**A**
16.	**A**	**A**	**B**
17.	**D**	**C**	**C**
18.	**C**	**B**	**B**
19.	**D**	**B**	**D**
20.	**A**	**B**	**C**
21.	**B**	**A**	**A**
22.	**B**	**D**	**C**
23.	**D**	**B**	**A**
24.	**C**	**C**	**A**
25.	**C**	**A**	**D**
26.	**A**	**D**	**C**
27.	**D**	**A**	**B**
28.	**C**	**D**	**C**
29.	**C**	**A**	**D**
30.	**B**	**B**	**B**

Physics

PART – I

SECTION – A

1. For maximum $\xrightarrow{1N}\xrightarrow{3N}\xrightarrow{9N}\xrightarrow{10N} \equiv \xrightarrow{23N}$

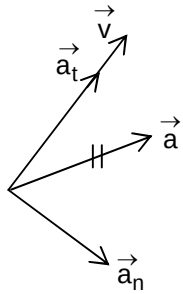


A triangle is possible with sides 4 unit, 9 unit and 10 unit. Hence minimum resultant is zero.

2. 2.4 km downstream from the point directly opposite to 0

3. rate of change of speed
= tangential component of acceleration
(component of acceleration in the direction of velocity)

$$= \vec{a}.\vec{v} = (2\hat{i}+\hat{j}).\left(\frac{3\hat{i}+3\hat{j}}{5}\right) = \frac{6+4}{5} = 2m/s^2$$

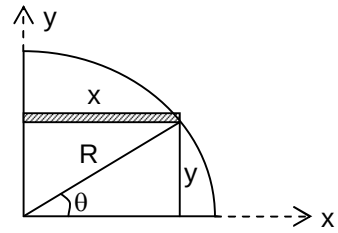


4. Force is always perpendicular to velocity

5. elementary mass = x dy
$x = R\cos\theta$, $y = R\sin\theta$
$dy = R\cos\theta\, d\theta$
elementary mass = $\sigma (R\cos\theta)(R\cos\theta\, d\theta)$
$= \sigma R^2\cos^2\theta\, d\theta$

$$y_{an} = \frac{\int y dm}{\int .dm} = \frac{\int_0^{\pi/2} R\cos\theta\, \sigma R^2\cos^2 d\theta}{\int_0^{\pi/2} \sigma R^2\cos^2\theta\, d\theta}$$

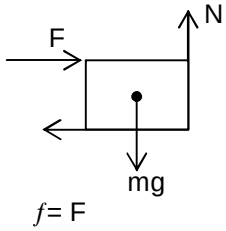
$$= \frac{4}{3\pi}R$$



6. Let the block is in translational equilibrium and ready to topple.
$f = F$
$N = mg$
Taking torque about C.O.M. = 0

$$F\left(\frac{a}{2}\right) + f\left(\frac{a}{2}\right) = N\left(\frac{a}{2}\right)$$

$2F = N$



$F = \frac{N}{2}$

$f = \frac{N}{2}$

but $f \le \mu N$

$\frac{N}{2} \le \mu N$

$\mu \ge \frac{1}{2}$

7. $f = f_o\left(\frac{V - V_o}{V - V_s}\right)$.

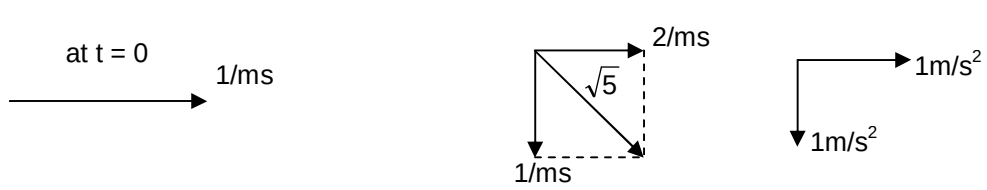
Here, $V_o = u + at$

$V_s = 0$

$\Rightarrow$ f varies linearly with 't'.

8. As long as cylinder is completely inside, B is constant in magnitude, then its value decreases linearly with displacement and finally becomes zero.

9.



$R = \frac{V^2}{a_n} = \frac{\left(\sqrt{5}\right)^2}{\frac{1}{\sqrt{5}}} = 5\sqrt{5}m$

10. Applying work-energy theorem w.r.t wedge

19. $V_P = \frac{kq_0}{2a} - \frac{kq_0}{a} + \frac{kq_0}{(a/2)} = \frac{3kq_0}{2a}$.

20. Pitch = $\frac{1.5}{5} = 0.3\,mm$

Least count = $\frac{0.3}{50} = 0.006\,mn$

21. $m_1v_1 = m_2v_2$

$\Rightarrow v_2 = m_1v_1 / m_2$ (1)

$\frac{1}{2}kx^2 - \mu m_2 g\frac{a}{2} = \frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2$ (2) From work energy theorem

Substituting (1) in (2) and solving

$v_1 = \sqrt{\frac{m_2(ka^2 - 4\mu m_2 ga)}{4m_1(m_1 + m_2)}}$ = 10 cm/s

22. Mechanical energy loss happens due to viscous/frictional forces without which the liquid will oscillate. This is released as heat

23. $mr\omega^2 = \dfrac{GMm}{r^3}$

$GM = r^3\omega^2$ $\qquad$ $(GM = gr^2)$

$g = \dfrac{r^3.\omega^2}{R^2}$

$g = \dfrac{(4nR)^3.\omega^2}{n^2.R^2}$

$g = 64\ nR\omega^2$.

24. Frequencies are in odd ratio hence it is a closed organ pipe.

$\text{freq} = (2n+1)\dfrac{v}{4L}$

$\Delta(\text{frequency}) = \dfrac{2v}{4L} = 100\text{Hz}$

$\Rightarrow$ L = 1.7 m.

25. For maximum elongation charges on the blocks must be equal to $\dfrac{Q}{2}$ on each block.

$\therefore \quad \dfrac{1}{4\pi\varepsilon_0}\dfrac{\left(\frac{Q}{2}\right)^2}{(\ell_0 + x)^2} = kx$

$Q = 2(\ell_0 + x)\sqrt{4\pi\varepsilon_0 kx}$.

26. Number of emitted electrons $= \dfrac{10^{12}\times 2\times 10^{-4}\times 25}{10^5}$ = 5 × 10^4 electrons

q = + ne = 8 × 10^{-15} C.

27. $R = \dfrac{3\times 6}{3+6}\Omega = 2\Omega$

$\dfrac{1}{R} = \dfrac{1}{R_1} + \dfrac{1}{R_2} \Rightarrow \dfrac{dR}{R^2} = \dfrac{dR_1}{R_1^2} + \dfrac{dR_2}{R_2^2}$

% error = $\dfrac{\Delta R}{R}\times 100\%$

$= R\left(\dfrac{\Delta R_1}{R_1}\times 100\times\dfrac{1}{R_1} + \dfrac{\Delta R_2}{R_2}\times 100\times\dfrac{1}{R_2}\right)\%$

$= 2\left(1\times\dfrac{1}{3} + 2\times\dfrac{1}{6}\right)\%$

$= \dfrac{4}{3}\%$

28. As V increases, E increases, λ_{min} decreases, $\lambda_{K\alpha}$ remains constant so ($\lambda_{K\alpha} - \lambda_{min}$) increases

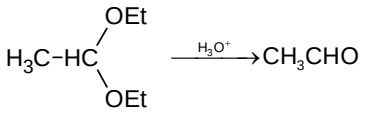
29. Increasing the filament current increases the intensity because more electron now strike the target. Decreasing the potential difference increases the minimum wavelength.

30. $\dfrac{(1.6-1)\times 1.8\times 10^{-5}}{900\times 10^{-9}}$ = 12

Chemistry

PART – II

SECTION – A

1. 
: differentiated by Na, Fehling and Tollen's test

(p) (Acetal)

$CH_3-CH_2-O-CH_2-CH_3 \xrightarrow{H_3O^+} EtOH$; differentiated by Na, Fehling and Tollen's test

(q)

2. $E^\circ_{X^-/AgX/Ag} = E^\circ_{Ag^+/Ag} + \frac{RT}{F}\ln Ksp$

3. Compounds with more covalent nature, was unable to respond to chromyl chloride test '$SnCl_4$', due to mole polarizing power of Sn^{+4} the compound is more covalent.

5. $K_c = \frac{[SO_3]^2}{[SO_2]^2[O_2]}$

$K_c = \frac{1}{[O_2]} = 100$

$[O_2] = 10^{-2}$

$\frac{n}{10} = 10^{-2}$

$\Rightarrow n = 0.1$

6. $n = n_1 + n_2 = \frac{P_1 2V}{T_1} = \frac{P_2 V}{T_1} + \frac{P_2 V}{T_2}$

8. $x_A = 0.3$, $x_B = 0.7$

$y_A = \frac{P_A}{P_T}, y_B = \frac{P_B}{P_T}$

$\frac{y_A}{y_B} = \frac{0.6}{0.4} = \frac{3}{2} = \frac{P^\circ_A x_A}{P^\circ_B x_B} = \frac{7}{2}$

9. RbCl and KCl have rock salt structure

$a_{RbCl} - a_{KCl} = 0.5$

$2(r_{Rb^+} - r_{Cl^-}) - 2(r_{K^+} - r_{Cl^-}) = 0.5$

Hence $r_{K^+} = 1.33\,\text{Å}$

11. Intramolecular aldol condensation.

12. Perkins's reaction

13. Bromination of benzylic carbon followed by hydrolysis

14. Hoffmann elimination, least stable alkene will be the major product.

16. β-Hydroxy compound is easily dehydrated.

17. $2AB_2(g) \rightleftharpoons 2AB(g) + B_2(g)$

1	0	0
1 - x	x	x/2

Total moles = 1 – x + x + x/2 = 1 + x/2.

$k_p = \frac{(x/2)(x)^2}{(1-x)^2}\left(\frac{p}{1+x/2}\right)$ [$\because$ x is small, 1-x = 1 and 1 + x/2 = 1].

$k_p = \frac{x^3 p}{2} \Rightarrow x = \sqrt{2k_p/P} = (2k_p/P)^{\frac{1}{3}}$

19. At constant pressure, the addition of inert gas shifts the equilibrium towards larger no. of moles.

20. The hybridisation of B in BBr_3 is sp^2 that C in CS_2 is sp, that of S in SO_2 is sp^2 while that of S in SF_4 is sp^3d. Hence CS_2 has the maximum bond angle of 180°.

27. $NaHCO_3 + NaOH \longrightarrow Na_2CO_3 + H_2O$

28. For a bcc lattice

$\frac{X^+}{Y^-} = 0.732 \Rightarrow Y^- = \frac{0.4}{0.732} = 0.546\,\text{Å}$

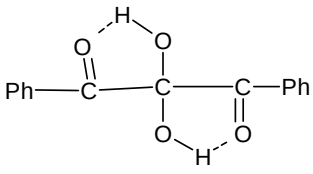
Now, for fcc lattice,

$\frac{Z^+}{Y^-} = 0.414 \Rightarrow Z^+ = Y^- \times 0.414$

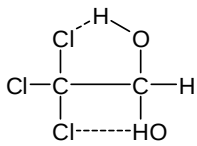
= 0.546 × 0.414

= 0.226 Å

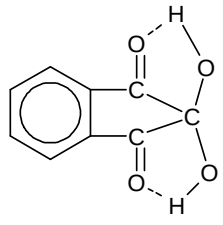
29.



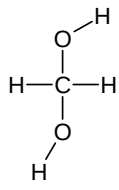
(I)



(II)



(III)



(V)

30. Hint: It is Acyl Cleavage

Mathematics

PART – III

SECTION – A

1. $f(x) = f(-x)$
$\Rightarrow x^n = -(-x)^n$

2. $\lim\limits_{x\to\infty} \dfrac{\log_a x}{x^a} \to 0$ and $\lim\limits_{x\to\infty} \dfrac{a^x}{\log_a x} \to \infty$

3. $g(x) = \cos(x) - ||x| - 1|,\ x \in R$

4. $\dfrac{dv}{dt} = \pi r\left(\dfrac{rdh}{dt} + \dfrac{2hdr}{dt}\right)$

5. $f(x) = \dfrac{ax^4}{4} + \dfrac{bx^3}{3} + \dfrac{cx^2}{2} + dx$
$f(0) = 0,\ f(-1) = \dfrac{a}{4} - \dfrac{b}{3} + \dfrac{c}{2} - d = 0$

6. $h(x) = f^2(x) + (f'(x))^2$
$h'(x) = -2x(f'(x))^2\, g(x)$

7. Let F be the fuel charges. Then, $F = \dfrac{3v^2}{16}$
Let train covers λ kms. Then, total cost for running the train, $C = \dfrac{3v\lambda}{16} + 300 \times \dfrac{\lambda}{v}$
For $\dfrac{dc}{dv} = 0$ and $\dfrac{d^2c}{dv^2} > 0$
$\Rightarrow$ v = 40 km/hr

8. For maximum or minimum, $f'(x) = 0$
$\Rightarrow -x^2 + 8x - (12 + t) = 0$
$\Rightarrow t < 4$

9. AD divides BC in ratio AB : AC
So, P.V. of D = $\dfrac{|\overrightarrow{AB}|(2\hat{i} + 5\hat{j} + 7\hat{k}) + |\overrightarrow{AC}|(2\hat{i} + 3\hat{j} + 4\hat{k})}{|\overrightarrow{AB}| + |\overrightarrow{AC}|}$

10. Required vector $\vec{r} = x_1\vec{b} + x_2\vec{c}$ and $\vec{r}\cdot\vec{a} = \sqrt{\dfrac{2}{3}}(|\vec{a}|)$
$\Rightarrow \vec{r} = -2\hat{i} + \hat{j}(x_1 - 2) - \hat{k}(-4 - x_1)$, where $x_1 \in R$

12. $\vec{r}\cdot\vec{a} = x_2[\vec{a}\ \vec{b}\ \vec{c}]$, $\vec{r}\cdot\vec{b} = x_3[\vec{b}\ \vec{c}\ \vec{a}]$, $\vec{r}\cdot\vec{c} = x_1[\vec{c}\ \vec{a}\ \vec{b}]$

13. Equation of line $x + 2y + z - 1 + \lambda(-x + y - 2z - 2) = 0$

$x + y - 2 + \mu(x + z - 2) = 0$
(0, 0, 1) lies on it
$\Rightarrow \lambda = 0, \mu = -2$
For point of intersection, $z = 0$

14. Equation of lines, $(y - mx) + \lambda, (z - c) = 0$
$(y + mx) + \lambda_2 (z + c) = 0$
It meets at x–axis, hence $\lambda_1 = \lambda_2$
$\Rightarrow cy = mzx$

15. Slope of reflected ray = $\frac{3}{4}$

$\Rightarrow$ Equation of incident ray is $(y + 4) = -\frac{3}{4}(x + 2)$ i.e., $4y + 3x + 22 = 0$

16. Required area = 4(Area of ΔS_2OS_4) = $4 \times \frac{1}{2} ae \times 8be_1$

17. $\sin^{-1}\left(\frac{\sqrt{r} - \sqrt{r-1}}{\sqrt{r(r+1)}}\right) = \tan^{-1}\left(\frac{\sqrt{r} - \sqrt{r-1}}{1 + \sqrt{r(r-1)}}\right) = \tan^{-1}\sqrt{r} - \tan^{-1}\sqrt{r-1}$

18. $AG = \frac{1}{3}\sqrt{2b^2 + 2c^2 - a^2} = \frac{1}{3}a$; $BG = \frac{1}{3}\sqrt{b^2 + 4c^2}$

$R_1 = \frac{(AG)(BG)(AB)}{4\Delta_{GAB}} = \frac{5\sqrt{13}}{12}$

19. $z^n - 1 = (z - 1)(z - z_1)(z - z_2) \ldots\ldots (z - z_{n-1})$
Differentiating w.r.t. x

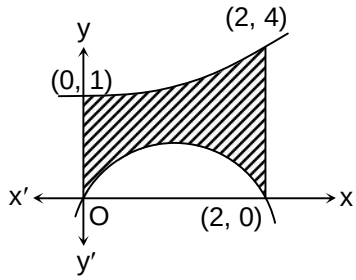
$$\frac{nz^{n-1}}{z^n - 1} = \frac{1}{z-1} + \frac{1}{z-2} + \ldots\ldots + \frac{1}{z - z_{n-1}}$$

20. $T_r = \frac{1}{2}\left(\frac{1}{1 \times 3 \times 5 \times \ldots\ldots \times (2r-1)} - \frac{1}{1 \times 3 \times 5 \times \ldots\ldots \times (2r+1)}\right)$

22. $\Delta = (\sin x + 2\cos x)\begin{vmatrix} 1 & \cos x & \cos x \\ 0 & \sin x - \cos x & 0 \\ 0 & 0 & \sin x - \cos x \end{vmatrix} = 0$

$\tan x = -2$ or $\tan x = 1$

23. Required area = $\int_0^2 \left(2^x - (2x - x^2)\right) dx$



26. Combined mean = 16; $d_1 = 9$; $d_2 = -6$

$$\sigma^2 = \frac{n_1\left(\sigma_1^2 + d_1^2\right) + n_2\left(\sigma_2^2 + d_2^2\right)}{n_1 + n_2} = 67.2$$

27. $1 \le 1 + \sin^3 x \le 2$

$$\frac{\pi}{2\sqrt{2}} \le I \le \frac{\pi}{2}$$

28. $x = e^{xy\frac{dy}{dx}}$

$$\Rightarrow \int y\,dy = \int \frac{\ln x}{x}\,dx$$

$$\Rightarrow y = \pm\sqrt{(\ln x)^2 + c}$$

29. Let A(α, 0) and B(β, 0) be the two points

$OT^2 = OA \cdot OB = \alpha\beta = \frac{c}{a}$

30. $f(x) = \frac{1 - x^{18}}{1 + x}$

$$\Rightarrow f(x-1) = \frac{1-(x-1)^{18}}{x}$$